

DARKE COUNTY AP, OH, USA

WMO: 720713

Lat: 40.204N

Lon: 84.532W

Elev: 1008

StdP: 14.17

Time Zone: -5.00 (NAE)

Period: 10-19

WBAN: 00251

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.5	6.8	-5.5	4.2	1.5	0.5	5.8	7.6	28.2	26.6	25.8	25.9	6.0	280	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	90.0	72.3	87.9	71.9	85.8	71.0	77.4	84.0	75.8	82.6	74.2	80.8	9.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.2	137.4	80.9	73.2	128.0	78.7	72.4	124.6	77.6	41.8	83.2	39.8	82.1	38.4	81.1	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.8	20.0	17.8	DB	-5.8	93.4	6.8	3.7	-10.7	96.1	-14.7	98.2	-18.5	100.2	-23.4	102.9	(4)
(5)			WB	-6.3	80.1	6.9	0.8	-11.2	80.7	-15.2	81.1	-19.0	81.6	-24.0	82.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.6	26.7	32.0	38.8	51.2	63.1	70.3	73.0	70.6	65.5	53.9	40.5	32.8		
	DBStd	18.55	12.19	11.65	10.90	8.75	9.06	5.99	5.44	4.85	7.76	9.30	9.26	9.98		
	HDD50	2633	724	515	381	89	11	0	0	0	0	65	309	539		
	HDD65	5714	1187	925	812	420	145	22	5	10	91	363	735	999		
	CDD50	3232	2	10	35	124	418	610	712	640	466	186	24	5		
	CDD65	839	0	0	1	5	87	183	253	185	106	19	0	0		
	CDH74	7302	0	0	10	69	753	1524	2200	1530	1034	179	3	0		
	CDH80	2224	0	0	0	4	197	473	732	408	372	38	0	0		
	WSAvg	7.4	9.8	9.4	9.2	9.2	7.5	6.6	4.5	3.9	4.9	7.2	8.2	8.5		
	PrecAvg	39.7	2.9	2.1	3.0	3.8	4.4	4.4	4.3	3.5	2.8	2.8	3.0	2.9		
(15) Precipitation	PrecMax	51.4	9.2	4.6	6.4	7.1	7.6	7.8	9.5	5.8	6.5	6.3	6.7	8.4		
	PrecMin	30.4	0.9	0.7	1.0	1.5	1.6	2.5	1.0	1.1	0.7	0.5	1.0	1.0		
	PrecStd	5.8	1.8	1.0	1.5	1.5	1.6	1.5	2.0	1.3	1.5	1.4	1.4	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.2	65.6	77.1	80.7	89.7	91.0	92.8	90.1	91.4	84.3	71.5	61.0		
		MCWB	56.6	55.9	62.4	61.9	69.1	71.7	74.3	72.8	71.1	66.5	61.7	58.1		
	2%	DB	53.9	59.4	68.0	75.4	85.5	88.1	89.8	87.6	88.3	79.1	64.1	55.3		
		MCWB	52.0	55.8	58.2	59.8	67.9	70.8	74.9	72.9	70.1	63.8	56.1	53.5		
	5%	DB	48.3	54.4	61.4	71.8	81.6	85.9	87.6	84.3	84.4	73.3	60.6	52.1		
		MCWB	47.0	51.7	55.1	58.2	66.6	69.4	73.8	72.3	69.2	62.5	53.4	50.7		
	10%	DB	43.5	48.3	55.4	66.3	78.8	81.9	84.2	81.7	80.8	70.0	54.9	45.9		
		MCWB	40.9	44.8	50.6	56.2	65.2	68.3	72.7	70.3	66.6	61.3	49.4	44.0		
	0.4%	WB	57.2	60.0	62.9	66.3	72.3	77.1	79.9	78.6	76.1	71.7	63.2	59.5		
		MCDB	59.9	63.6	72.5	76.1	79.7	85.3	85.3	83.9	83.7	78.7	66.2	60.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	52.1	55.7	59.7	63.0	70.9	75.0	78.1	76.7	73.9	68.5	59.0	54.4		
		MCDB	52.8	56.8	66.9	69.5	78.8	82.2	84.3	82.6	82.3	73.5	63.5	55.9		
	5%	WB	46.4	51.5	56.1	60.7	69.2	73.2	76.6	74.9	72.1	65.6	54.2	50.2		
		MCDB	47.4	55.7	60.5	66.9	77.1	80.4	83.4	80.8	79.4	71.8	58.3	51.7		
	10%	WB	41.3	45.0	52.4	58.6	67.8	71.7	74.7	73.1	70.1	62.5	50.2	44.8		
		MCDB	43.3	47.3	55.3	65.3	76.4	78.5	81.4	78.7	76.3	68.0	55.3	45.7		
	5% DB	MDBR	13.9	14.7	15.9	21.0	19.9	19.4	19.5	21.0	22.6	20.3	16.3	11.9		
		MCDBR	17.9	21.6	22.1	27.3	23.5	23.4	21.2	23.4	26.7	25.4	23.5	17.2		
		MCWBR	16.3	17.6	15.1	14.8	9.8	9.2	9.1	10.8	10.0	12.4	15.5	15.6		
		MCWBR	17.0	20.7	19.8	21.5	19.4	18.0	17.8	18.2	21.8	20.0	20.4	16.5		
(40) Clear-Sky Solar Irradiance	taub		0.301	0.320	0.344	0.400	0.441	0.442	0.446	0.440	0.399	0.354	0.324	0.299		
		taud	2.418	2.351	2.363	2.218	2.146	2.197	2.198	2.198	2.285	2.404	2.445	2.455		
	Ebn at Noon		274	283	288	277	267	266	264	262	266	267	262	266		
		Edh at Noon	26	32	35	43	48	46	45	44	38	30	25	23		
(44) All-Sky Solar Radiation	RadAvg		528	816	1129	1509	1735	1926	1909	1736	1411	978	646	444		
	RadStd		48	90	95	132	147	137	109	83	123	89	80	48		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+189	N/A

Nomenclature: See separate page